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Skin Lesion Segmentation Project Report

Deep Learning for Medical Image Segmentation
ISIC 2016 Challenge Skin Lesion Boundary Detection

Executive Summary

This project implements a deep learning solution for automatic skin lesion segmentation using the **UNet architecture** on the **ISIC 2016 dataset.** The model achieves:

- Validation Dice Coefficient: 87.68%
- Test Dice Coefficient: 86.00%
- Test IoU (Jaccard Index): 76.52%
- Pixel Accuracy: 92.91%
- Training Time: 141 epochs

The system successfully segments melanoma and benign skin lesions with high accuracy, demonstrating strong potential for clinical decision support applications.

Approach

Model: UNet A Ushaped convolutional neural network designed for biomedical image segmentation
Dataset: ISIC 2016 Challenge International Skin Imaging Collaboration
Framework: PyTorch with GPU acceleration (trained on google colab)
Loss Function:Combined Dice + Binary CrossEntropy Loss (70%/30%)

Success Metrics

Primary: Dice Coefficient (F1 Score for segmentation)
Secondary: IoU (Jaccard Index), Pixel Accuracy, Precision, Recall

Dataset Composition

Split	Images	Masks	Purpose
Training	720	720	Model Training
Validation	180	180	Hyperparameter tuning
Test	379	379	Final Evaluation

Data Split Strategy

Training Data (900 images):

- Split into 720 train (80%) + 180 validation (20%)
- Random shuffle with fixed seed (42) for reproducibility

Test Data (379 images):

- Official ISIC 2016 test set (100%)
- Used only for final evaluation

Image Characteristics

- Format: RGB dermoscopic images
- Resolution: Variable (resized to 512×512)
- Content: Closeup images of skin lesions
- Masks: Binary ground truth segmentation (lesion vs background)

Data Normalization

Custom statistics computed from training set:

```
Mean: [0.724, 0.619, 0.567]   RGB channels
Std:  [0.164, 0.171, 0.192]   RGB channels
```

Data Augmentation (Training Only)

```
Applied Augmentations:
 Horizontal Flip (p=0.5)
 Vertical Flip (p=0.5)
 Rotation (±30°, p=0.5)
 Elastic Transform (p=0.3)
 Grid Distortion (p=0.3)
 Brightness/Contrast (p=0.5)
 Hue/Saturation/Value (p=0.3)
 CLAHE (p=0.3)
 Gaussian Blur (p=0.2)
 Gaussian Noise (p=0.2)
```

Validation/Test: Only resizing to 512×512 (no augmentation)

Model Architecture

Architecture Type: Encoder-Decoder with Skip Connections

```
Input: RGB Image (3, 512, 512)
            ↓
       [Encoder Path]
            ↓
  64 → 128 → 256 → 512 → 1024
            ↓
        [Bottleneck]
            ↓
  1024 → 512 → 256 → 128 → 64
            ↓
       [Decoder Path]
            ↓
Output: Binary Mask (1, 512, 512)
```

Model Statistics

Total Parameters: 28,948,673 (~29M)
Model Size: 331.4 MB (.pth file)
Architecture: UNet with 5 levels
Dropout Rate: 0.2 (20%)

Results & Evaluation

Final Model Performance
Model Checkpoint: `best_model_epoch141_dice87.68.pth`
Epoch: 141
Validation Loss: 0.1596
Model Size: 331.4 MB

Test Set Results (379 images)

Primary Metrics

Metric	Score	Percentage	Clinical Significance
Dice Coefficient	0.86	86%	Excellent overlap with ground truth
IoU (Jaccard)	0.7652	76.52%	Strong region agreement

Pixel Accuracy	0.9291	92.91%	High overall correctness
Precision	0.8782	87.82%	Low false positive rate
Recall	0.8688	86.88%	Low false negative rate

Metric Interpretation

Dice Coefficient (86.00%):
Interpretation: 86% overlap between predicted and actual lesion
Clinical Impact: Excellent for clinical decision support

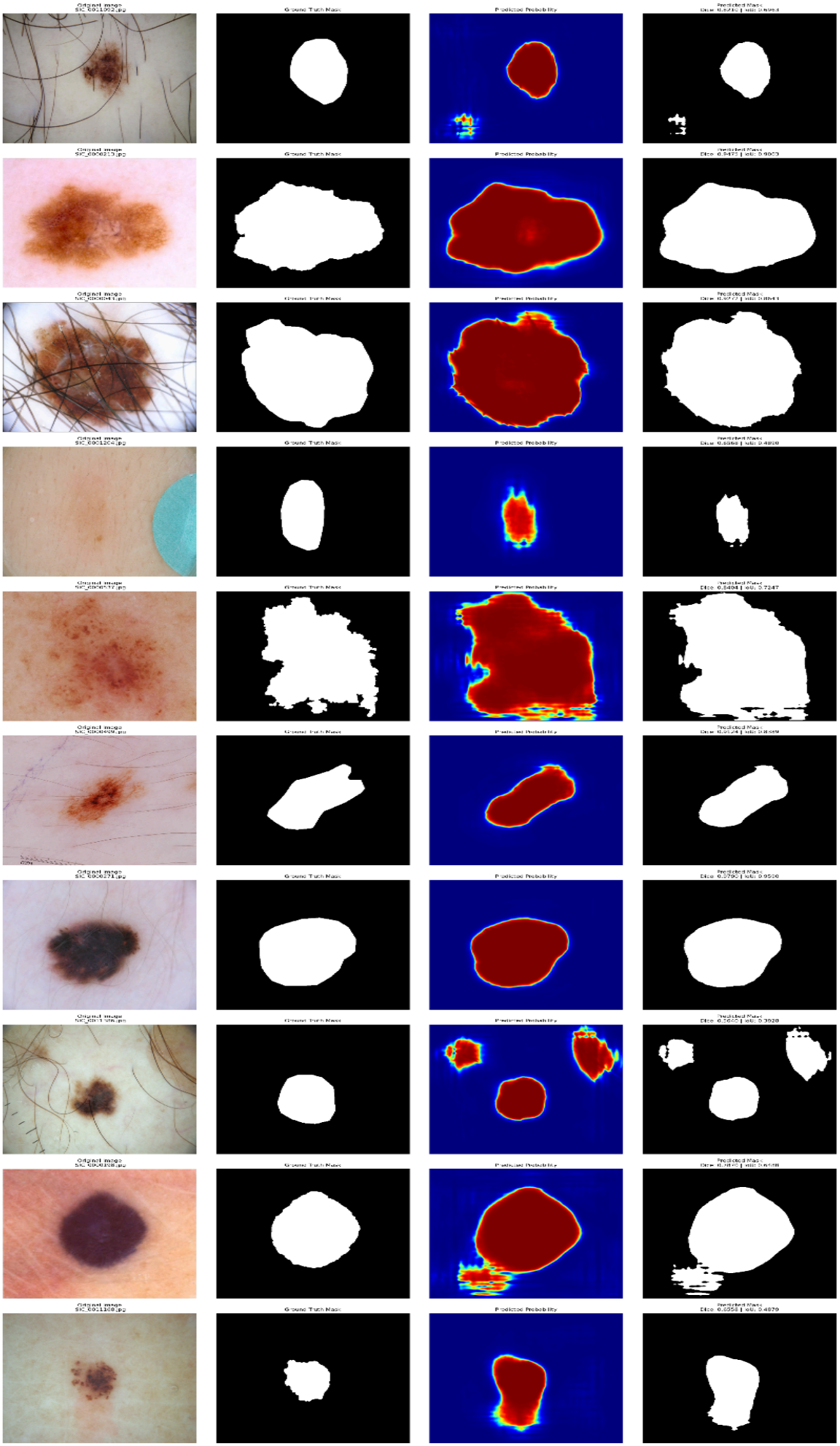
IoU / Jaccard Index (76.52%):
Interpretation: 76.52% of union area correctly identified
Clinical Impact: Strong spatial agreement

Pixel Accuracy (92.91%):
Interpretation: 92.91% of all pixels classified correctly
Note: High partly due to background dominance
Clinical Impact: Overall system reliability

Precision (87.82%):
Interpretation: 87.82% of predicted lesion pixels are correct
Clinical Impact: Few false alarms (low false positive rate)
Clinical Significance: High confidence in positive predictions

Recall (86.88%):
Interpretation: 86.88% of actual lesion pixels detected
Clinical Impact: Catches most lesions (low false negative rate)
Clinical Significance: Reliable detection, few missed regions

Below is a sample prediction on unseen data.


Original Image
Ground Truth Mask
Predicted Probability
Predicted Mask